6TiSCH and Routing in Low-Power Networks (RPL, CoRPL, CARP)

Communication Technologies for the Internet of Things, Lecture 10

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Agenda (1/2)

- Networks for critical scenarios: the motivation for 6TiSCH
- Limitations of the classic IEEE 802.15.4 MAC
- IEEE 802.15.4e Time-Slotted Channel Hopping
- Frequency hopping and transmission scheduling
- 6TiSCH reference architecture
- Routing in low-power and lossy networks: RPL

Agenda (2/2)

- Variants and alternatives: CoRPL and CARP
- Application scenarios

What 6TiSCH Is (1/2)

- It corresponds to an implementation of IEEE 802.15.4 aimed at critical scenarios.
- In industrial environments, delivery criticality has two simultaneous requirements: the message must be delivered, and within short and precise time bounds.
- This demand is radically different from what is asked of an environmental sensor network, where a lost or delayed reading is tolerable.
- The 6TiSCH working group aims to enable IPv6 over the Time-Slotted Channel Hopping mode of the IEEE 802.15.4e standard.

What 6TiSCH Is (2/2)

- It targets low-power wireless devices, often called *motes*, organised in a multi-hop low-power and lossy network.
- These networks connect to the conventional Internet through one or more border routers, offering guarantees for critical traffic.

Why Bring IP to Industrial Networks (1/2)

- IP became the norm in information technology and on the Internet, with all the advantages that follow.
- There is an abundance of people with IP skills, of diagnostic tools, of cheap hardware and of software, libraries and utilities.
- In contrast, industrial networks are expensive: hardware is costly, technicians to install and manage them are scarce, and there are few competing implementations.

Why Bring IP to Industrial Networks (2/2)

- This contrast motivated the move to reuse IP-related technologies inside industrial networks.
- It is, in essence, the same argument that underpinned 6LoWPAN, now applied to a domain with much stricter timing requirements.

The Obstacle: Soft Guarantees versus Hard Guarantees (1/2)

- IP networks work on the basis of buffers that guarantee the network service.
- Information technology and Internet services live well with delay variation (*jitter*) and network latency.
- IP quality-of-service approaches only offer soft guarantees: buffering packets lets applications minimise network disturbances, but does not eliminate them.
- Industrial and automotive traffic requires hard delivery guarantees, with deadlines in the order of milliseconds or microseconds.

The Obstacle: Soft Guarantees versus Hard Guarantees (2/2)

- It also requires network redundancy, with recovery in equally short intervals.
- It is this incompatibility between the IP model and industrial requirements that 6TiSCH sets out to resolve.

Alternative Approaches (1/2)

- **WirelessHART (2007) and ISA100.11a:** born inside the industrial environment and remained there, with no significant adoption outside it.
- **Audio Video Bridging (AVB):** approach promoted by the IEEE.
- **Time Sensitive Networking (TSN):** successor of AVB, also within the IEEE.
- The last two were driven by automotive scenarios, with a strong focus on combining leisure traffic with control traffic on the same medium.

Alternative Approaches (2/2)

- An important distinction: AVB and TSN are approaches for wired networks.
- None of these solutions, by itself, solves the problem of bringing the advantages of the IP ecosystem to low-power wireless networks with strict timing requirements.

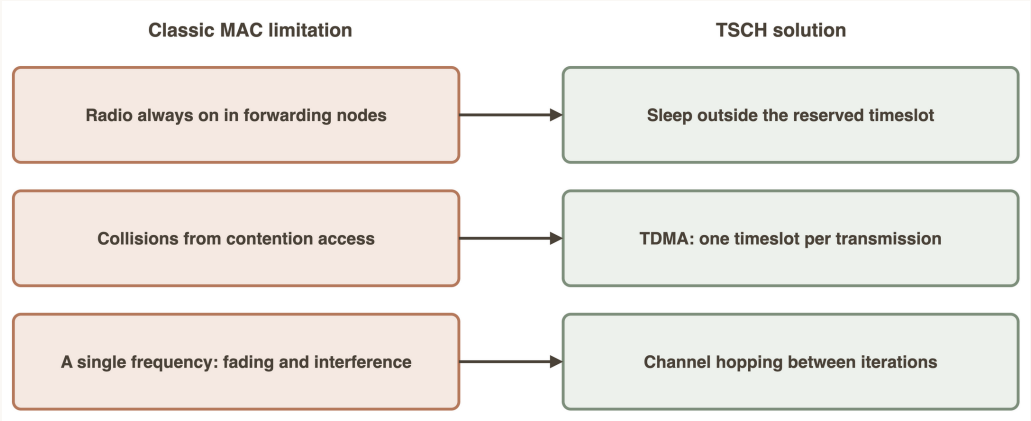
Limitations of the Classic IEEE 802.15.4 MAC (1/2)

- High energy consumption in forwarding nodes, which have to keep the radio permanently on, regardless of the traffic they actually carry.
- Additional consumption from failed transmission attempts, when they occur.
- Collisions, inherent to contention-based medium access.
- A high level of interference and fading, because all communication takes place on a single frequency.

Limitations of the Classic IEEE 802.15.4 MAC (2/2)

- The IEEE 802.15.4e standard was published in 2012, as an amendment to the MAC of IEEE 802.15.4-2011, to address these limitations.
- It introduces a deterministic low-latency network, with channel hopping in time slots and deterministic, synchronous multichannel extension.
- The amendment only affects the MAC layer: it does not change the physical layer, keeping operation at 2.4 GHz.

From Problem to TSCH Solution



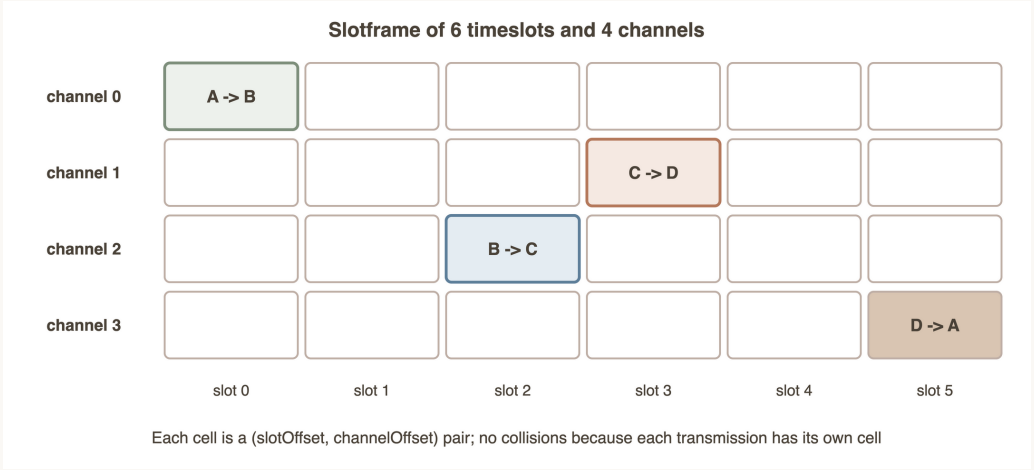
Medium Access in TSCH (1/2)

- Time intervals (*timeslots*) are grouped into one or more slotframes, which repeat cyclically.
- Each node in the network transmits within its own slot, in a TDMA-based scheme.
- Slot scheduling can be distributed or centralised, depending on the architecture adopted.
- Since each transmission takes place in a reserved slot, there are no collisions on the network.

Medium Access in TSCH (2/2)

- Stations can go to sleep after transmitting or receiving, which saves substantial energy compared with the classic model.
- Synchronisation between nodes is ensured by Enhanced Beacons, without which the TDMA scheme would not work.

Slotframe and Cells



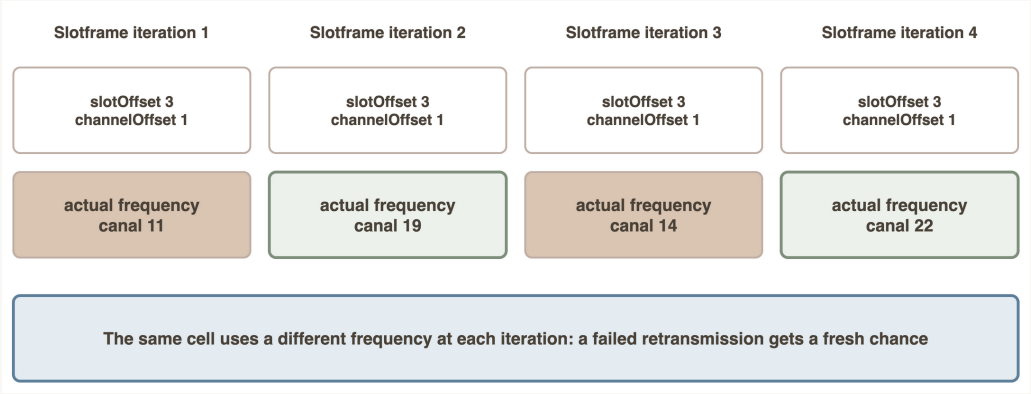
Frequency Hopping (1/2)

- Neighbouring nodes using a cell obtain a different frequency at the next iteration of the slotframe.
- This happens even when the cell keeps the same slot offset (*slotOffset*) and the same channel offset (*channelOffset*).
- Wireless propagation phenomena, such as multipath fading and external interference, affect a radio link in a frequency-dependent way.
- After a failed transmission, retransmitting on a different frequency has a higher probability of success than retransmitting on the same frequency.

Frequency Hopping (2/2)

- Channel hopping therefore produces more stable links and, as a consequence, a more stable network topology.
- This stability is precisely what makes it possible to offer the timing guarantees that critical scenarios require.

One Cell, Several Frequencies



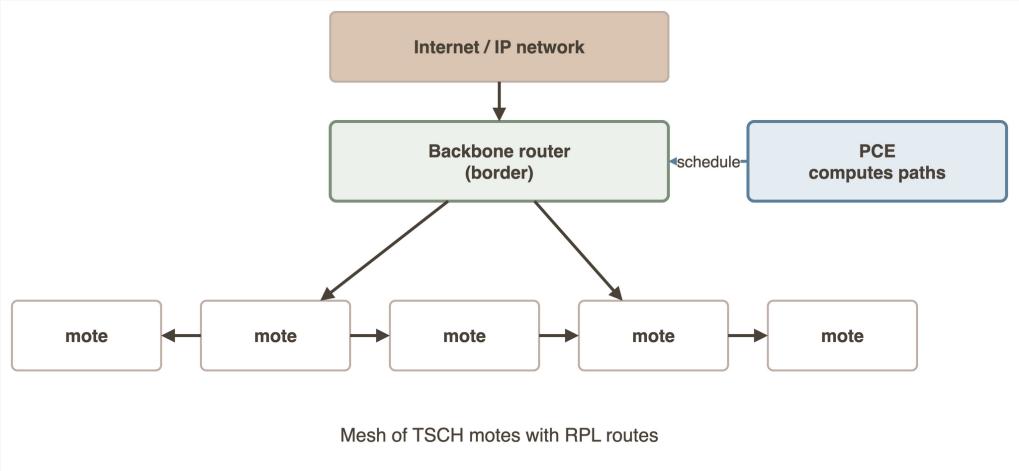
6TiSCH Network Architecture (1/2)

- Nodes are organised in a mesh network, with multiple possible paths between points.
- Backbone routers connect the sensor network to the information technology network and allow high-throughput connectivity between subnets of nodes.
- There is a management entity, the Path Computation Element, responsible for computing and assigning the paths that traffic follows through the network nodes.

6TiSCH Network Architecture (2/2)

- This separation between data plane and control plane is what allows centralised scheduling.
- Forwarding over 6TiSCH is done over routes defined by the RPL protocol, which links this architecture to the routing covered in the second part of the lecture.

The 6TiSCH Architecture



Transmission Scheduling and Standards (1/2)

- **Centralised scheduling:** uses protocols such as PCEP, ForCES or OpenFlow, in which a central entity determines the schedule of all nodes.
- **Distributed scheduling:** uses protocols such as RSVP or NSIS, over a previously established network, with nodes negotiating resources among themselves.
- The working group seeks to reuse existing standards, namely IPv6 Neighbor Discovery, 6LoWPAN addressing and the RPL routing protocol.

Transmission Scheduling and Standards (2/2)

- **Published standards:** RFC 7554, on the use of TSCH; RFC 8180, on the minimal configuration of a 6TiSCH network; and RFC 8480, which defines the 6top protocol.
- The 6TiSCH architecture has also been standardised by the IETF.

Routing in Low-Power and Lossy Networks (1/2)

- Low-Power and Lossy Networks (LLNs) have characteristics that rule out conventional routing protocols.
- Nodes have very limited memory, processing and energy, which excludes protocols that require extensive routing tables.
- Links are unstable and asymmetric, with loss rates that vary over time.
- Traffic is mostly directed to a single collection point, and not evenly distributed between pairs of nodes.

Routing in Low-Power and Lossy Networks (2/2)

- Protocols such as RIP or OSPF, seen in Lecture 08, assume stable links and nodes with resources, assumptions that do not hold here.
- It is this gap that the protocols covered next try to fill.

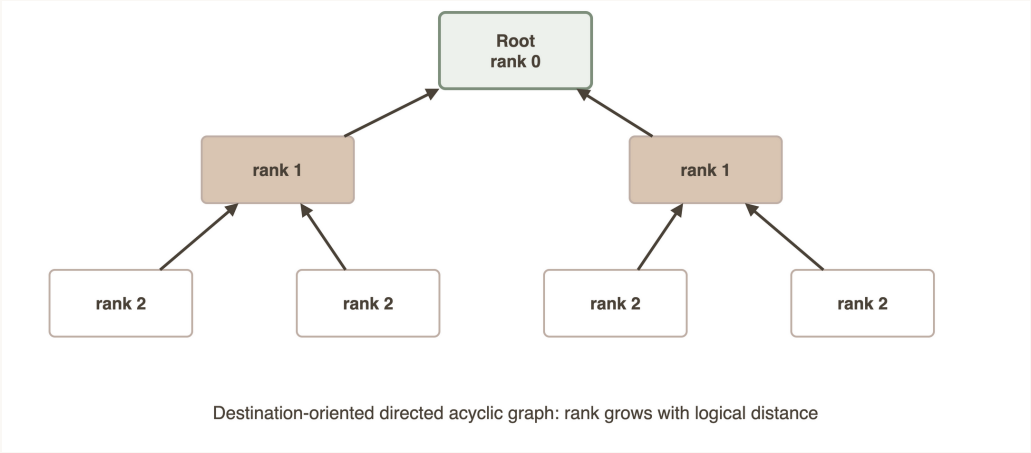
RPL: Concept and DODAG Topology (1/2)

- RPL (*IPv6 Routing Protocol for Low-Power and Lossy Networks*) is defined in RFC 6550 and is the reference routing protocol for these networks.
- It builds a logical topology called a DODAG (*Destination-Oriented Directed Acyclic Graph*): a directed, acyclic graph oriented towards a destination.
- The root of the graph is typically the data collection node, that is, the border router that connects the network to the Internet.
- The absence of cycles is guaranteed by construction, which avoids circular routing problems without the need for costly detection mechanisms.

RPL: Concept and DODAG Topology (2/2)

- Each node gets a *rank* value, which expresses its logical distance from the root and establishes the hierarchy of the graph.
- The *rank* increases as nodes move away from the root, and it is the comparison of ranks that prevents a node from choosing one of its own descendants as parent.

A DODAG with Ranks



Destination-oriented directed acyclic graph: rank grows with logical distance

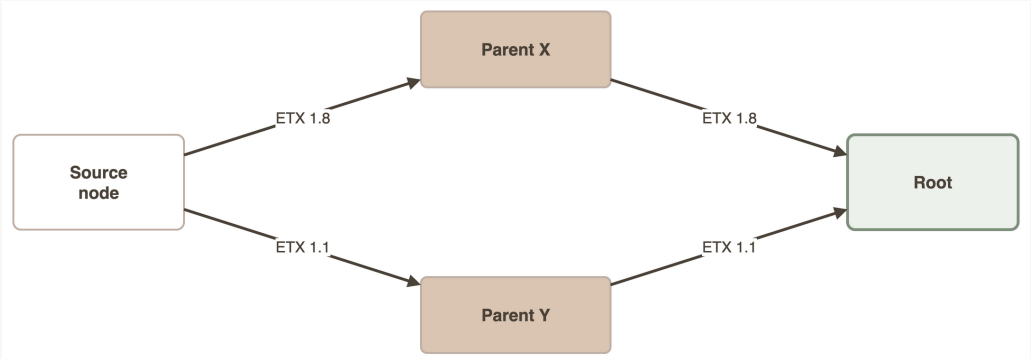
RPL: Objective Function and Metrics (1/2)

- The Objective Function determines how each node computes its rank and how it selects its preferred parent.
- This separation between the protocol mechanism and the optimisation criterion is a deliberate feature of RPL: it allows the same protocol to be adapted to different goals.
- An Objective Function may favour the minimum number of hops, a simple criterion but one that is insensitive to link quality.

RPL: Objective Function and Metrics (2/2)

- It may alternatively favour link quality, measured by metrics such as ETX (*Expected Transmission Count*), which estimates the number of transmissions needed to deliver a packet successfully.
- It may also consider the residual energy of nodes, avoiding exhausting those with less autonomy.
- The choice of Objective Function has a direct impact on network lifetime and delivery reliability.

Choosing the Parent: Hops or ETX



RPL: Control Messages and Trickle Timer (1/3)

- **DIO (*DODAG Information Object*):** broadcast by nodes to announce the existence of the graph and its parameters, letting new nodes join and choose their parent.
- **DAO (*Destination Advertisement Object*):** sent towards the root to announce reachable destinations, supporting downward traffic.
- **DIS (*DODAG Information Solicitation*):** used by a node to request the emission of DIO messages, when it wants to join without waiting for the periodic broadcast.

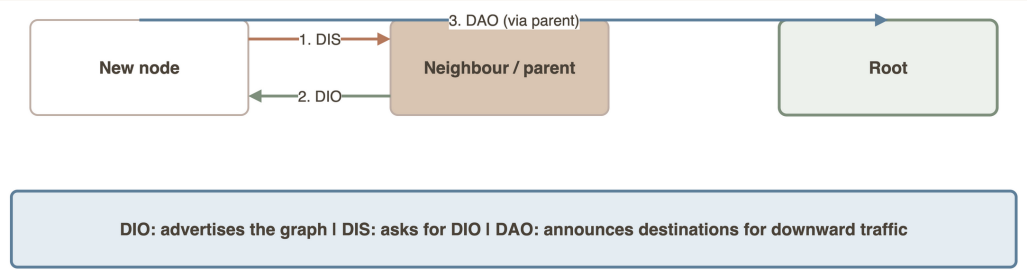
RPL: Control Messages and Trickle Timer (2/3)

- RPL distinguishes multipoint-to-point traffic, from the nodes to the root, and point-to-multipoint, from the root to the nodes, also supporting point-to-point communication through the root.
- The *Trickle Timer* adaptively controls the frequency of control messages: when the network is stable, the interval between emissions grows progressively, reducing control traffic and consumption.

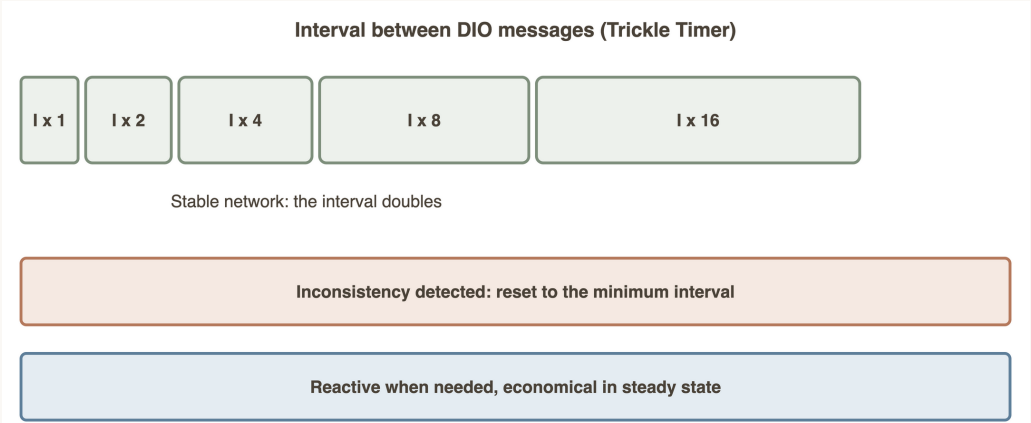
RPL: Control Messages and Trickle Timer (3/3)

- When an inconsistency is detected, the timer is reset to its minimum value, which speeds up the propagation of the new information.
- This mechanism is what reconciles reactivity to change with energy economy in steady state.

DIS, DIO and DAO



The Trickle Timer



CoRPL (1/2)

- CoRPL is a variant of RPL aimed at scenarios with variable channel conditions, associated with cognitive and opportunistic networks.
- It keeps the DODAG structure of RPL, but changes the way the next hop is determined.
- Instead of fixing a single preferred parent node, CoRPL uses sets of candidate forwarders (*forwarder sets*).
- Next-hop selection is opportunistic: among the candidates that actually received the transmission, the one in the best condition to continue forwarding is chosen.

CoRPL (2/2)

- This approach takes advantage of the broadcast nature of the radio medium, in which a transmission is potentially heard by several neighbours at once.
- The gain is greater robustness on unstable links, since the failure of one candidate does not imply retransmission if another received the message.

CARP (1/2)

- CARP (*Channel-Aware Routing Protocol*) is a distributed routing protocol, originally proposed for underwater sensor networks, but applicable to other low-power and lossy networks.
- Its distinctive feature is to use the link-quality history, observed in earlier transmissions, as the criterion for selecting the next hop.
- A node that has shown consistent delivery success is preferred over a node with an irregular history, even if geographically more favourable.
- The protocol does not depend on the extensive exchange of global topology information, which reduces control traffic and the memory needed in each node.

CARP (2/2)

- It also considers the residual energy of nodes, avoiding overloading those with less autonomy.
- It is particularly suited to environments where propagation conditions vary sharply and unpredictably.

Comparing the Three Protocols (1/2)

- **RPL:** DODAG structure with a fixed preferred parent, selection criterion defined by a configurable Objective Function, and adaptive control of signalling traffic through the Trickle Timer. It is the IETF-standardised protocol and the most widely implemented.
- **CoRPL:** keeps the RPL structure, but replaces the single parent with sets of candidates and opportunistic selection. It gains robustness on unstable channels, at the cost of greater complexity in coordinating candidates.
- **CARP:** dispenses with the global hierarchical structure and decides the next hop based on link-quality history and available energy. It reduces control traffic, but offers fewer guarantees about the global structure of routes.

Comparing the Three Protocols (2/2)

- The choice between them depends on the expected stability of the medium and on the degree of standardisation and interoperability the solution requires.
- In installations that must interoperate with third-party equipment, RPL is the natural choice, being the standardised protocol.

RPL, CoRPL and CARP Side by Side

RPL	CoRPL	CARP
Fixed preferred parent	Set of candidate forwarders	No global structure
Configurable Objective Function	Opportunistic selection	Link quality history
IETF standard	More robust, more complex	Less control traffic

Application Scenarios and Summary (1/2)

- **6TiSCH application scenarios:** home automation, vehicular scenarios including in-vehicle communications, and industrial communications.
- 6TiSCH answers a concrete problem: the classic IEEE 802.15.4 MAC forces forwarding nodes to keep the radio always on, suffers collisions and concentrates all communication on a single frequency.
- TSCH resolves these three limitations through TDMA access with reserved slots, channel hopping between iterations and synchronisation through enhanced beacons.

Application Scenarios and Summary (2/2)

- The absence of collisions and the stability of links are what make it possible to offer hard timing guarantees, unlike the soft guarantees typical of IP networks.
- On this basis, RPL provides routing, organising the network in a destination-oriented directed acyclic graph, with configurable selection criteria and adaptive signalling control.
- CoRPL and CARP represent alternatives aimed at more unstable media, favouring respectively opportunistic selection and link-quality history.

Thank you.

Questions and comments